Design Questions - B

1. Online Library Management System:

The system allows users to search for books, borrow books, return books, and manage their accounts. Librarians can add/remove books, manage user accounts, and generate reports. The system also sends notifications for overdue books.

Questions:

- Draw a Level 0 DFD (Context Diagram) for the Online Library Management System, showing the interaction between the system, users, and librarians.
- Create a Level 1 DFD that breaks down the "Borrow Book" process, including interactions with the database and notifications.
- Create a Sequence Diagram for the "Borrow Book" use case, showing the interactions between the User, System, Database, and Notification Service.
- Design a Class Diagram for the system, including classes like User, Librarian, Book, Transaction, and Notification. Show relationships such as inheritance, association, and aggregation.
- Write detailed use cases for "Borrow Book" and "Return Book". Include actors, preconditions, postconditions, and the flow of events.

2. Online Food Delivery System

System Description: The system allows customers to browse restaurants, place orders, track deliveries, and make payments. Restaurant owners can manage menus and orders. Delivery personnel can view and update delivery statuses.

- Draw a Level 0 DFD for the Online Food Delivery System, showing interactions between customers, restaurants, delivery personnel, and the system.
- Create a Level 1 DFD for the "Place Order" process, including interactions with the payment gateway and restaurant.
- Create a Sequence Diagram for the "Track Delivery" use case, showing interactions between the Customer, System, Delivery Personnel, and Database.
- Design a Class Diagram for the system, including classes like Customer, Restaurant, Order, Delivery, and Payment. Show relationships such as composition, aggregation, and dependency.
- Write detailed use cases for "Place Order" and "Track Delivery". Include actors, preconditions, postconditions, and the flow of events.

3. Hospital Management System

System Description: The system allows patients to book appointments, view medical records, and receive prescriptions. Doctors can manage appointments, update patient records, and prescribe medications. Administrators can manage staff and generate reports.

- Draw a Level 0 DFD for the Hospital Management System, showing interactions between patients, doctors, administrators, and the system.
- Create a Level 1 DFD for the "Book Appointment" process, including interactions with the doctor's schedule and patient records.
- Create a Sequence Diagram for the "Update Medical Record" use case, showing interactions between the Doctor, System, Database, and Patient.

- Design a Class Diagram for the system, including classes like Patient, Doctor, Appointment, MedicalRecord, and Prescription. Show relationships such as inheritance, association, and composition.
- Write detailed use cases for "Book Appointment" and "Update Medical Record". Include actors, preconditions, postconditions, and the flow of events.

4. E-Commerce Platform

System Description: The system allows users to browse products, add items to a cart, place orders, and make payments. Admins can manage products, orders, and user accounts. The system also sends order confirmation emails.

- Draw a Level 0 DFD for the E-Commerce Platform, showing interactions between users, admins, payment gateways, and the system.
- Create a Level 1 DFD for the "Place Order" process, including interactions with the payment gateway and inventory system.
- Create a Sequence Diagram for the "Add to Cart" use case, showing interactions between the User, System, Database, and Inventory.

- Design a Class Diagram for the system, including classes like User, Product, Order, Cart, and Payment. Show relationships such as aggregation, composition, and dependency.
- Write detailed use cases for "Add to Cart" and "Place Order". Include actors, preconditions, postconditions, and the flow of events.

5. Online Banking System

System Description: The system allows customers to check balances, transfer funds, pay bills, and view transaction history. Admins can manage user accounts and generate reports. The system also sends transaction notifications.

- Draw a Level 0 DFD for the Online Banking System, showing interactions between customers, admins, and the system.
- Create a Level 1 DFD for the "Transfer Funds" process, including interactions with the account database and notification service.
- Create a Sequence Diagram for the "Pay Bill" use case, showing interactions between the Customer, System, Database, and Bill Payment Service.
- Design a Class Diagram for the system, including classes like Customer, Account, Transaction, Bill, and Notification. Show relationships such as inheritance, association, and aggregation.
- Write detailed use cases for "Transfer Funds" and "Pay Bill". Include actors, preconditions, postconditions, and the flow of events.

6. University Course Registration System

System Description: The system allows students to register for courses, view schedules, and check grades. Professors can manage courses and submit grades. Admins can manage student and course records.

- Draw a Level 0 DFD for the University Course Registration System, showing interactions between students, professors, admins, and the system.
- Create a Level 1 DFD for the "Register for Course" process, including interactions with the course database and student records.

- Create a Sequence Diagram for the "Submit Grade" use case, showing interactions between the Professor, System, Database, and Student.
- Design a Class Diagram for the system, including classes like Student, Professor, Course, Registration, and Grade. Show relationships such as inheritance, association, and composition.
- Write detailed use cases for "Register for Course" and "Submit Grade". Include actors, preconditions, postconditions, and the flow of events.

7. Microservices Architecture: E-Commerce Platform

System Description: You are tasked with designing a scalable e-commerce platform using microservices architecture. The platform should support product browsing, order management, payment processing, and user notifications. Each service should be independently deployable and scalable.

- How would you decompose the e-commerce platform into microservices? Provide a list of potential services (e.g., Product Service, Order Service, Payment Service) and justify your decomposition strategy.
- What criteria would you use to define service boundaries to ensure loose coupling and high cohesion?

8. IoT and Edge-Based Architecture: Smart City Traffic Management

System Description: You are designing an IoT and edge-based architecture for a smart city traffic management system. The system collects data from traffic sensors, cameras, and GPS devices, processes it at the edge, and provides real-time traffic updates and optimizations (e.g., dynamic traffic light control).

- Why is edge computing critical for this system, and how would you balance edge and cloud processing? Provide examples of tasks that should be performed at the edge versus the cloud.
- What are the trade-offs between latency, bandwidth, and computational power in this architecture?

9. Hybrid Architecture: Healthcare Monitoring System

System Description: You are designing a hybrid architecture for a healthcare monitoring system that combines microservices, serverless, and edge computing. The system collects patient data from wearable devices, processes it at the edge for real-time alerts, and stores it in the cloud for long-term analysis.

- How would you decompose the system into microservices (e.g., Patient Data Service, Alert Service, Analytics Service)? What are the key considerations for designing healthcare-related microservices?

10. Draw the cyclomatic complexity graph and determine the number of test cases required

```
def calculate_grade(score):
    if score >= 90:
        grade = 'A'
    elif score >= 80:
        grade = 'B'
    elif score >= 70:
        grade = 'C'
```

```
    else:
        grade = 'F'  # Node 7
    return grade
```

1. Draw the Control Flow Graph (CFG) for the given code. Label each node (e.g., Node 1, Node 2, etc.) and edges to represent the flow of execution.
2. Calculate the cyclomatic complexity of the code using the formula:
   $V(G)=E-N+2V(G)=E-N+2$, where:
   - EE = Number of edges
   - NN = Number of nodes
3. Determine the number of test cases required to achieve full path coverage based on the cyclomatic complexity

11. Function Point Analysis Problem

Problem Statement:

You are tasked with estimating the size of a Library Management System using Function Point Analysis. The system has the following components:

1. External Inputs (EI):
   - Add New Book: 3 inputs
   - Issue Book: 2 inputs
   - Return Book: 2 inputs
   - Search Book: 1 input
2. External Outputs (EO):
   - Generate Report: 2 outputs
   - Display Book Details: 1 output
3. External Inquiries (EQ):
   - Check Book Availability: 1 inquiry
   - View User History: 1 inquiry
4. Internal Logical Files (ILF):
   - Book Database: 1 file
   - User Database: 1 file
5. External Interface Files (EIF):
   - External Library API: 1 file

1. Determine Complexity:
   - Low Complexity: EI = 3, EO = 4, EQ = 3, ILF = 7, EIF = 5
   - Average Complexity: EI = 4, EO = 5, EQ = 4, ILF = 10, EIF = 7
   - High Complexity: EI = 6, EO = 7, EQ = 6, ILF = 15, EIF = 10

   *Assume all components are of Average Complexity.
2. Calculate Unadjusted Function Points (UFP):
3. Determine Value Adjustment Factor (VAF):

*. Assume the Total Degree of Influence is 42 (based on 14 general system characteristics, each rated on a scale of 0 to 5).